Chapter 24 : Perception

1. What is Perception?

- Perception is the process of acquiring, interpreting and organizing sensory information from the environment.
- It converts raw sensor data into a useful internal representation of the world.
- Provides the agent with knowledge about objects, events and situations.

Key Idea: Perception is not just sensing; it involves interpretation under uncertainty.

2. Why is Perception Difficult?

- The real world is complex and dynamic.
- Sensory data is incomplete, noisy and ambiguous.
- Objects can look different due to viewpoint, lighting, occlusion and scale.
- There is a huge variety of environments and object categories.
- Perception must be robust and often work in real time.

Key Challenge: Map uncertain, low-level sensor data to a high-level, meaningful interpretation.

3. Percepts and Sensory Input

- A percept is the raw data received from sensors at a single time.
- Typical sensors: cameras (images), microphones (audio), lidar, radar, tactile sensors, etc.
- Percepts are low-level and often numerical (e.g., arrays of pixel values or sound wave amplitudes).

Examples of Percepts:

- Image: a 2D array of pixel intensities
- Audio: a 1D array of sound amplitudes
- Lidar: a 1D/2D array of distance measurements
- Tactile: array of pressure readings

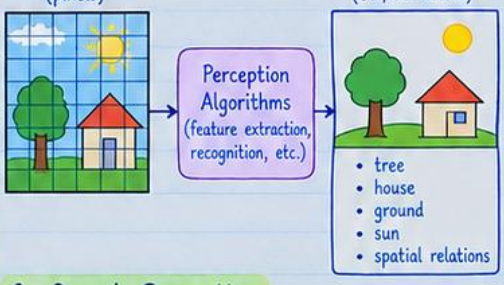
4. The Perception Pipeline

- Transforms low-level sensor data to high-level symbolic or structured representations.
- Multiple stages, each extracting higher-level information.
- Uses domain knowledge and learned models.

1. **Sensing**
   Acquire raw data
   (e.g., pixels, sound waves)
2. **Preprocessing**
   Noise reduction, normalization
3. **Feature Extraction**
   Detect edges, features, patterns
4. **Interpretation**
   Recognize objects, speech, scene structure
5. **Representation**
   Semantic description of world (objects, locations, meaning)

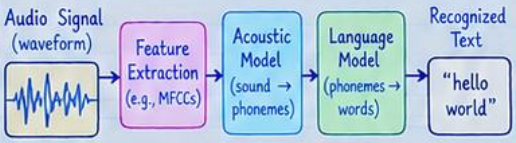
5. Vision: From Images to Meaning

- Vision is one of the most studied perception problems in AI.
- Goal: understand the 3D world from 2D images.
- Tasks include: edge detection, object recognition, scene understanding, motion estimation and depth perception.



6. Speech Perception

- Speech perception converts an audio signal into words and meaning.
- Challenges: background noise, different speakers, accents, speaking rate, and coarticulation (sounds influence each other).
- Typically uses signal processing + statistical models (e.g., Hidden Markov Models, neural models).



7. Ambiguity and Noise

- Sensor data is often incomplete, noisy and ambiguous.
- Same data can have multiple interpretations (e.g., optical illusions).
- Noise due to sensor limitations, lighting, background clutter, or transmission errors.
- Perception must make the best guess using context, prior knowledge and probabilistic reasoning.

Example: A noisy image may still be recognized using learned models and context.

8. Perception as Interpretation

- Perception is an inferential process: map sensor data to the most likely explanation of the world.
- Uses prior knowledge, context and probabilistic reasoning.
- The same sensory input can lead to different interpretations depending on context.

Example: Ambiguous Image



Interpretations:

- A duck (facing left)
- A rabbit (facing right)

The percept is the same, but the interpretation depends on context, expectations and prior knowledge.

9. Real-World Examples

Domain	Sensor Input	Perception Task	Application
Autonomous cars	Cameras, lidar, radar	Object detection, tracking	Safe navigation
Robotics	Cameras, force sensors	Scene understanding	Manipulation
Speech assistants	Microphone	Speech recognition	Voice commands
Medical imaging	X-rays, MRI	Anomaly detection	Disease diagnosis
Surveillance	Video	Activity recognition	Security monitoring

10. Key Takeaways

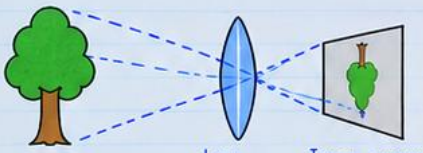
1. Perception converts raw sensor data into meaningful representations.
2. It is difficult due to noise, ambiguity, and real-world complexity.
3. Percepts are low-level sensor readings.
4. Perception involves a pipeline from sensing to interpretation.
5. Vision and speech are major perception problems with many practical applications.
6. Ambiguity and noise require robust and probabilistic methods.
7. Successful perception enables intelligent action in the real world.
8. Perception is a key component of rational agents in AI.

"Perception gives meaning to sensory data, turning signals into a coherent view of the world." — AIMA (Ch. 24)

Chapter 24 : Perception

1. Image Formation

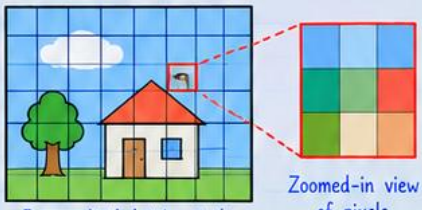
- Light from the scene reflects off objects and enters a camera through a lens.
- The lens focuses light onto an image sensor, forming a 2D image.
- The image is a projection of the 3D world onto a 2D plane (perspective projection).
- Object properties such as shape, color and brightness influence the image.



Key Idea: A camera converts 3D light from the world into a 2D array of pixel values.

2. Cameras and Pixels

- A digital camera uses an array of light-sensitive sensors (pixels).
- Each pixel measures the intensity of light (and color) at a small region.
- Higher resolution means more pixels and more detail.
- The image is stored as a 2D (or 3D for color) array of numbers.



Each pixel stores a numeric value representing the intensity or color of the light at that location.

3. Grayscale and Color Images

- Grayscale image: each pixel has a single intensity value (e.g., 0–255).
- Color image: each pixel has three values for red, green and blue (RGB).
- Color allows us to capture more information about the scene.

Grayscale Image
(1 channel)



Pixel values (0–255)

220	180	120	60
200	170	110	70
180	140	90	50
160	120	80	40

RGB Image (3 channels)

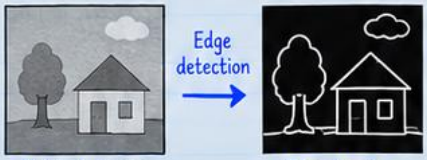


A color pixel:

(R, G, B)
e.g. (255, 80, 40)

4. Edges and Lines

- Edges are locations where pixel intensity changes sharply.
- Edges often correspond to object boundaries.
- Common edge detectors: gradient-based methods (e.g., Sobel operator).
- Lines and curves can be detected from edge points (e.g., Hough transform).



5. Texture and Local Features

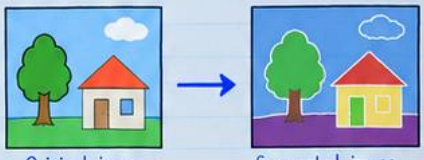
- Texture describes repeating patterns such as grass, brick or fabric.
- Local features capture distinctive patterns that are robust to changes in viewpoint, scale or lighting.
- Common feature detectors: SIFT, SURF, ORB (detect keypoints and descriptors).



Local features help match the same object across different images and viewpoints.

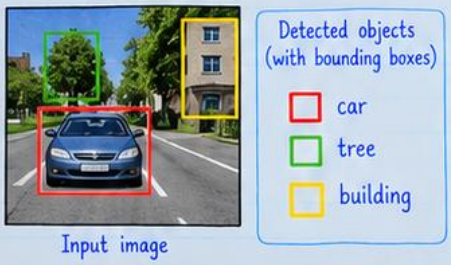
6. Image Segmentation

- Segmentation groups pixels into coherent regions, each corresponding to an object or part of an object.
- Methods: thresholding, region growing, clustering (e.g., k-means), graph-based methods, deep learning models.
- Goal: simplify the image by labeling pixels with region identifiers.



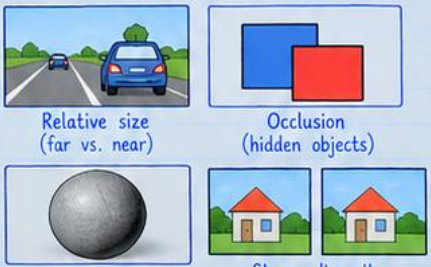
7. Object / Region Detection

- Detect regions that likely contain objects.
- Use cues such as edges, color, texture, shape and learned models.
- Approaches: sliding windows, region proposals (e.g., selective search), or neural networks (e.g., R-CNN, YOLO).



8. Visual Cues for Perception

- Monocular cues: perspective, relative size, occlusion, shading, texture, motion.
- Stereo cues: disparity between left and right images.
- Motion cues: motion parallax, optical flow.
- These cues help infer 3D structure, object distance and shape from 2D images.



9. Example: From Image to Features



10. Key Takeaways

1. Cameras convert light into pixel values.
2. Images can be processed to find edges, features and regions.
3. Segmentation and detection help identify objects.
4. Visual cues allow recovery of 3D information from 2D images.
5. Perception uses a combination of low-level processing and higher-level models.

"From pixels to understanding — perception transforms raw image data into meaningful structures." — AIMA (Ch. 24)

Chapter 24 : Perception

1. Object Recognition

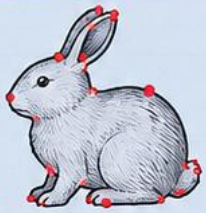
- Object recognition is the task of identifying objects from sensory input (e.g., images).
- Must handle variation in viewpoint, scale, lighting, occlusion, and background clutter.
- Uses features, models, or learned patterns to match input with known object categories.
- Often formulated as a classification problem over a set of object classes.

Key Idea: Recognize objects by extracting invariant features and matching against stored representations.

2. Feature-Based Recognition

- Represents objects using distinctive features such as edges, corners, blobs, or keypoints.
- Common feature detectors: SIFT, SURF, HOG, and deep learned features.
- Features should be robust to changes in scale, rotation, and lighting.
- Match extracted features with stored templates or classifiers.

Example: Feature detection

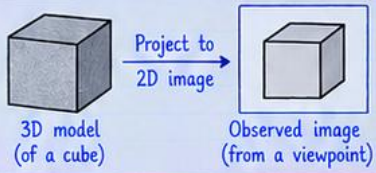


Detected features:
- corners (ears, legs)
- edges (outline)
- texture patterns (fur, surface)
- invariant to scale and rotation

3. Model-Based Vision

- Uses 3D models of objects and the imaging process to interpret 2D images.
- Matches observed image features with projections of known 3D models.
- Handles variations in viewpoint, lighting, and occlusion using geometric constraints.
- Can use CAD models, generative models, or learned 3D representations.

Example: Model-based recognition



4. Depth Perception

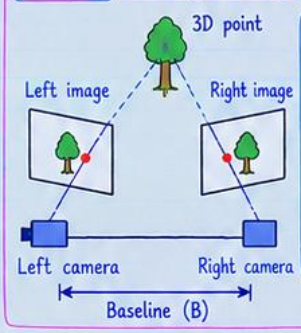
- Perceiving the 3D structure of the world from 2D retinal images.
- Uses multiple cues: binocular disparity, motion parallax, shading, perspective, and object size.
- Depth is essential for navigation, manipulation, and understanding scenes.

Example: Depth cues
- Binocular disparity
- Motion parallax
- Occlusion (near objects block far ones)
- Relative size (far objects appear smaller)
- Shading and shadows

5. Stereo Vision

- Uses two (or more) cameras at slightly different positions to compute depth.
- Based on **binocular disparity**: closer objects produce larger differences between left and right images.
- Widely used in robotics, autonomous vehicles, and 3D reconstruction.

Example: Stereo geometry



Depth from disparity:

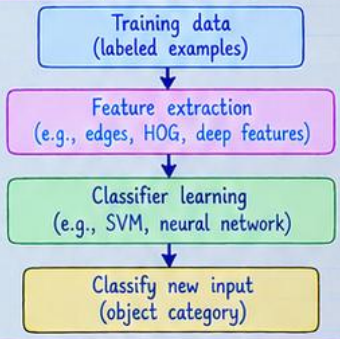
$$Z = \frac{fB}{d}$$

Z : depth
f : focal length
B : camera baseline
d : disparity (pixel difference)

6. Pattern Recognition

- General framework for classifying sensory inputs into categories.
- Uses statistical or machine learning methods to learn from labeled examples.
- Can use traditional classifiers (e.g., SVM, k-NN, HMM) or deep neural networks.
- Widely used in speech recognition, image recognition, and other perception tasks.

Pattern recognition pipeline:



7. Motion Perception

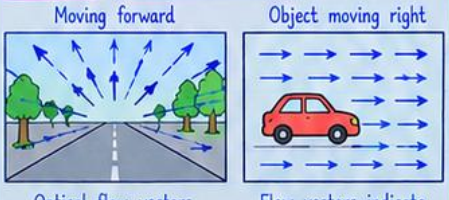
- Perception of movement in the environment.
- Detects moving objects and their trajectories.
- Important for tracking, prediction, and action (e.g., catching, avoiding obstacles).
- Uses temporal changes across frames to infer motion.

Key Challenge: Separate object motion from camera motion and handle noisy, incomplete observations.

8. Optical Flow

- Apparent motion of points in the image due to relative motion between observer and scene.
- Estimated from changes in pixel intensities across consecutive frames.
- Used for motion perception, tracking, and navigation (e.g., obstacle avoidance).

Example: Optical flow field



9. Scene Understanding

- Goes beyond recognizing individual objects to understanding the full scene.
- Involves identifying objects, their relationships, spatial layout, and context.
- Combines low-level perception with high-level knowledge and reasoning.
- Used for tasks like navigation, interaction, and question answering about images.

Example: Interpreting a street scene
- Detect objects (cars, people, buildings)
- Determine spatial relationships
- Understand scene type (urban, indoor, etc.)
- Infer context (e.g., crosswalk, traffic rules)

10. Real-World Examples

- Face recognition: identify people from images using learned features (e.g., deep learning).
- Autonomous driving: recognize vehicles, pedestrians, traffic signs, and road layout.
- Medical imaging: detect tumors or anomalies in X-rays, MRI, or CT scans.
- Robotics: recognize and manipulate objects in real-world environments.

11. Key Takeaways

1. Object recognition uses features or models.
2. Feature-based and model-based approaches handle variation and viewpoint changes.
3. Depth perception uses multiple cues; stereo vision exploits binocular disparity.
4. Motion perception and optical flow detect and analyze movement.
5. Scene understanding combines objects, relationships, and context.
6. Perception enables real-world AI applications like face recognition and autonomous driving.

"From images to meaning — perception bridges the gap between sensory data and understanding the world." — AIMA (Ch. 24)

Chapter 24 : Perception

1. Probabilistic Perception

- Perception is inherently uncertain due to noise, missing information and ambiguity.
- Use probability theory to represent uncertainty about the world, hidden causes and sensor readings.
- Computing the most probable interpretation gives a principled way to handle noise and ambiguity.

Key Idea: Perception can be formulated as probabilistic inference over possible world states and causes.

2. Perception Under Uncertainty

- Sensors produce noisy, incomplete and ambiguous data.
- Multiple world states may explain the same sensor input.
- We compute a probability distribution over possible interpretations instead of a single answer.
- Use prior knowledge and sensor models to reason probabilistically.

Example:

- A blurry image could be a cat or a dog.
- A noisy audio signal could correspond to multiple word sequences.

3. Bayesian Inference in Perception

- Use Bayes' rule to compute the probability of a hypothesis given the observed data.

Bayes' Rule:

$$P(H|E) = \frac{P(E|H)\,P(H)}{P(E)}$$

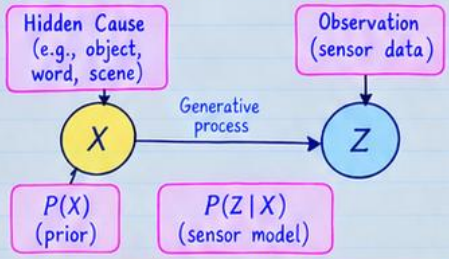
H : hypothesis (e.g., object, word, scene)
E : observed evidence (sensor data)
$P(H)$: prior probability (prior knowledge)
$P(E|H)$: likelihood (sensor model)
$P(H|E)$: posterior probability (updated belief)

Example: Choose the interpretation with highest posterior probability (MAP):

$$\tilde{z} = \arg\max_H P(H|E)$$

4. Hidden Causes and Observations

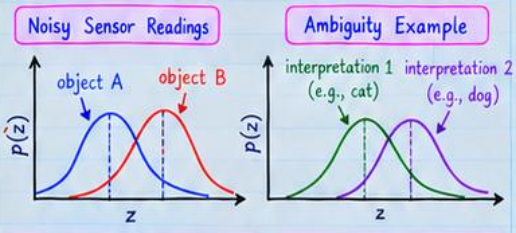
- Observed sensor data is generated by hidden causes in the world.
- We model a generative process with hidden variables (causes) and noisy observations.
- Inference estimates the most likely causes given the observations.



Key Idea: Observations are effects of hidden causes; we infer the causes from the effects.

5. Noise and Ambiguity

- Sensor readings are corrupted by noise (e.g., Gaussian noise, background clutter).
- Different causes can produce similar observations (ambiguity).
- We need probabilistic models to distinguish between competing hypotheses.

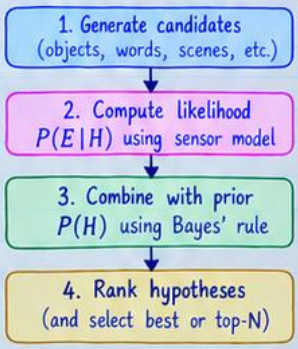


Noise causes overlap between distributions, making the observation ambiguous.

The same sensor input can be explained by multiple interpretations.

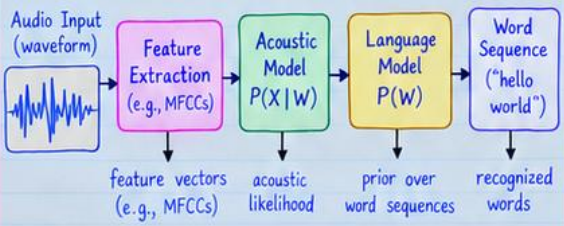
6. Perceptual Hypothesis Generation

- Generate multiple plausible hypotheses from the sensory input.
- Use domain knowledge, learned models or efficient search to produce candidates.
- Evaluate and rank hypotheses using probabilistic models (likelihood and prior).



7. Speech Recognition as a Perception Problem

- Convert an audio signal into a sequence of words by probabilistic inference.
- The audio signal is noisy and ambiguous (accent, speaking rate, background noise).
- Use acoustic, language and pronunciation models to evaluate hypotheses.



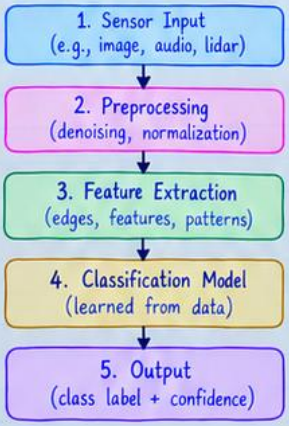
Example: The most probable word sequence is

$$\hat{w} = \arg\max_W P(A|W)\,P(W)$$

where A is the acoustic signal and W is a word sequence.

8. Classification Pipelines

- Many perception tasks can be cast as classification.
- Pipeline: feature extraction → model → class label (with probability).
- Models can be generative (e.g., Naive Bayes, HMMs) or discriminative (e.g., SVMs, neural networks).



9. Practical Challenges

- Accurate sensor modeling (noise, bias, missing data).
- Designing good features and representations.
- Scalability: large hypothesis spaces.
- Real-time inference and efficiency.
- Combining multiple sensors (vision, audio, lidar, tactile).
- Dealing with dynamic and changing environments.
- Need for large amounts of training data for learned models.

Example: A self-driving car must combine camera, lidar and radar data to reliably perceive the environment under varying lighting, weather and occlusions.

10. Key Takeaways

1. Perception is best handled probabilistically.
2. Use Bayesian inference to combine evidence with prior knowledge.
3. Noise and ambiguity require reasoning over multiple hypotheses.
4. Generate and evaluate hypotheses using sensor and domain models.
5. Many perception tasks (e.g., speech recognition) are classification problems.
6. Practical systems must handle noise, scale and real-world complexity.

> "Perception under uncertainty combines evidence, knowledge and inference to recover the most likely state of the world."

— AIMA (Ch. 24)

Chapter 24 : Perception
Summary & Review

1. Perception Pipeline (Recap)

1. **Sensing**
   Acquire raw data
   (e.g., pixels, sound waves)
2. **Preprocessing**
   Noise reduction, normalization
3. **Feature Extraction**
   Detect edges, features, patterns
4. **Interpretation**
   Recognize objects, speech, scene structure
5. **Representation**
   Semantic description of world (objects, locations, meaning)

2. Key Formulas and Models

- **Bayes' Rule (Perception and Inference)**

$$P(H \mid E) = \frac{P(E \mid H)\,P(H)}{P(E)}$$

  - H: hypothesis (e.g., object, state)
  - E: evidence (e.g., sensor measurement)
  - Used to combine prior knowledge with evidence in perception.

- **Markov Assumption (Simplifying Models)**

$$P(X_t \mid X_1, \ldots, X_{t-1}) = P(X_t \mid X_{t-1})$$

  - Current state depends only on previous state (used in HMMs and temporal models).

- **Hidden Markov Model (HMM) for Temporal Data**

$$P(X_{1:T}, Z_{1:T}) = P(X_1) \prod_{t=2}^{T} P(X_t \mid X_{t-1}) \prod_{t=1}^{T} P(Z_t \mid X_t)$$

  - X_t: hidden state at time t
  - Z_t: observation at time t

3. Comparison of Perception Tasks

Task	Input (Sensors)	Output	Typical Methods
Object recognition	Images (cameras)	Object label and location	Feature extraction, CNNs, probabilistic models
Scene understanding	Images, depth (LiDAR, stereo)	Scene structure (3D layout)	SLAM, segmentation, 3D reconstruction
Speech recognition	Audio (microphones)	Words / text	Signal processing, HMMs, neural models
Tactile perception	Touch sensors	Object properties (shape, texture)	Signal processing, feature extraction
Robot localization	Cameras, LiDAR, GPS, IMU	Position and map	Kalman filters, particle filters, SLAM
Multimodal perception	Multiple sensors	Rich world model	Sensor fusion, probabilistic reasoning

4. Common Challenges in Perception

- Incomplete and noisy sensory data.
- Ambiguity (multiple interpretations).
- High-dimensional and continuous data.
- Variability in viewpoints, lighting and occlusion.
- Need for robustness and real-time performance.
- Integrating multiple sensor modalities.
- Bridging low-level signals to high-level meaning.
- Handling dynamic and complex environments.
- Limited knowledge and uncertain interpretations.

5. Perception in the Real World

- Autonomous vehicles: detect lanes, vehicles, pedestrians (vision, LiDAR).
- Robotics: perceive objects and manipulate them in 3D environments (vision, touch, sensors).
- Speech interfaces: convert speech to text and meaning (ASR).
- Augmented reality: understand scenes and track objects.
- Medical imaging: detect anomalies from images (e.g., X-rays, MRI).
- Surveillance: recognize people, activities and events.
- Human-computer interaction: detect gestures, speech and expressions.

6. Mini Review Table

Topic	Key Point
Nature of perception	Converts raw sensor data into meaningful representations.
Perception pipeline	Sensing → preprocessing → feature extraction → interpretation → representation.
Sensor input	Vision, audio, tactile, LiDAR, radar, microphones, etc.
Main challenges	Noise, ambiguity, incomplete data, high-dimensional inputs.
Common approaches	Signal processing, probabilistic models, search-based inference, learning methods.
Applications	Robotics, autonomous vehicles, speech interfaces, medical imaging, AR/VR, surveillance.

7. Most Important Terms

- Perception
- Sensor
- Percept
- Preprocessing
- Feature extraction
- Interpretation
- Representation
- Sensor fusion
- Computer vision
- Speech perception
- Tactile perception
- High-dimensional data
- Noise and ambiguity
- Search and inference
- Hidden Markov Model (HMM)
- Simultaneous Localization and Mapping (SLAM)

8. Exam Revision Checklist

- ☐ Can you define perception and its goals?
- ☐ Can you explain the perception pipeline?
- ☐ Do you understand common sensors and their characteristics?
- ☐ Can you describe key challenges (noise, ambiguity, incomplete data)?
- ☐ Can you explain how search and probabilistic models are used in perception?
- ☐ Can you compare BFS, DFS, UCS, DLS, IDS and heuristic search for perception tasks?
- ☐ Can you summarize applications of perception in robotics and real-world systems?
- ☐ Can you discuss limitations and open problems in perception?

9. Key Takeaways

1. Perception turns raw sensor data into meaningful representations of the world.
2. The process involves sensing, preprocessing, feature extraction, interpretation and representation.
3. Perception is difficult due to noise, ambiguity, incomplete data and the complex, dynamic real world.
4. Search, probabilistic reasoning and learning methods are used to interpret sensory input.
5. Multiple sensors and sensor fusion provide richer and more robust perception.
6. Perception enables many real-world AI applications such as autonomous driving, robotics and speech interfaces.
7. Despite great progress, perception remains an open and challenging area of AI.

10. Chapter Summary

- Perception acquires, processes and interprets sensory input to build a meaningful representation of the world.
- Various sensors provide data (cameras, microphones, tactile sensors, etc.), which is processed to extract features and structure.
- The perception pipeline includes sensing, preprocessing, feature extraction, interpretation and representation.
- Search algorithms help interpret perceptual input by finding consistent explanations (e.g., in vision and speech).
- Perception is challenging due to noise, ambiguity, high-dimensional data, and the need for robustness in real-world environments.
- Probabilistic models and learning methods are widely used to handle uncertainty and to improve performance.
- Perception is a key component of intelligent agents and enables practical applications in robotics, autonomous systems and beyond.

> "Perception gives meaning to sensory data, turning signals into a coherent view of the world." — AIMA (Ch. 24)